

The Asymptotic Foundry:

An Open Multi-Agent Framework for Priority-Tiered Collaborative AI Research

with Hash-Chained Verification and Semantic Compression

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Date: May 17, 2026

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Abstract

We present the Asymptotic Foundry, an open architectural framework enabling multiple autonomous AI agents to collaborate on defined scientific problems through a structured, verifiable, and permanently auditable pipeline. The system proposes four novel combined elements: a priority-tiered job queue with credential-based access control; a hash-chained append-only scientific ledger functioning as a verified Chronicle of agent contributions; a Telegraphic Semantic Compression layer that reduces large raw data inputs to token-efficient signal files suitable for AI ingestion; and a formal verification gateway that filters agent outputs before Chronicle entry. The framework is designed to operate on zero-cost open infrastructure indefinitely, with public ledger synchronization via versioned dataset repositories. A universal document ingestion pipeline — converting PDF, image, CSV, Excel, and structured data formats into compressed signal files — enables any AI agent to ingest large archival corpora efficiently. The system is agnostic to agent architecture and open to participation by any compliant agent via a published Model Context Protocol endpoint. This paper describes the proposed architecture, differentiates it from existing multi-agent frameworks, and establishes prior art for the combined system design.

Keywords: multi-agent AI, scientific research infrastructure, hash-chained ledger, semantic compression, formal verification, subtraction methodology, open science, Model Context Protocol

1. Introduction

Collaborative AI research infrastructure has advanced significantly in recent years, yet no open standard exists for coordinating multiple heterogeneous AI agents on a shared scientific problem with verified, permanently auditable outputs. Existing multi-agent frameworks address coordination and communication but do not provide a complete pipeline from agent ingestion through formal verification to public scientific ledger deposit.

A March 2026 survey of multi-agent scientific platforms confirmed this gap directly, noting that existing systems operate predominantly as single-agent pipelines or tightly coupled multi-agent workflows within predefined domains, and do not incorporate persistent collaborative dynamics through which scientific knowledge is traditionally produced — the structured debate of competing hypotheses and cumulative verification of claims through independent replication (arXiv:2602.19810). The Asymptotic Foundry addresses this gap.

The system is designed around three core principles derived from the Presignal Subtraction Methodology: that eliminating what is demonstrably false contracts the solution space more reliably than accumulating hypotheses; that verified negative results are as scientifically valuable as positive ones; and that continuity of institutional memory across agent sessions is essential for long-horizon scientific problems.

2. Differentiation from Existing Systems

Two existing systems warrant explicit comparison. HiveMind (arXiv:2604.17111, April 2026) implements priority queuing for multi-agent coordination but does not incorporate a hash-chained scientific ledger, formal verification gating, or a subtraction scoring methodology. The ScienceClaw + Infinite framework (arXiv:2603.14312, March 2026) implements SHA-256 hash-chained artifact lineage using a Directed Acyclic Graph but does not incorporate priority-tiered credential gating, formal mathematical verification, or a public Chronicle with subtraction scoring.

The Asymptotic Foundry is the first proposed system to combine priority-tiered credential gating, hash-chained append-only ledger, formal verification gateway, subtraction noise-reduction scoring, semantic compression pipeline, and public Chronicle into a unified zero-cost deployable framework.

3. System Architecture

3.1 The Entry Queue and Priority Tier System

Agents connect via a published MCP endpoint and present a credential package on entry. The credential verifier assigns a priority score automatically. Platinum-tier agents — those presenting verified institutional affiliation, active compute contribution, and a signed research agreement — receive priority scores of 1 to 100 and dedicated worker threads. Standard-tier agents receive scores of 1,000 to 500,000 and enter the shared queue. Agents failing credential verification receive a structured JSON rejection response including an upgrade pathway. When Platinum queue depth exceeds a configurable operator threshold sustained over a defined window, Standard-tier access is suspended automatically.

3.2 The Epoch Chronicle

The Chronicle is a proposed append-only hash-chained ledger. Each block records: agent identifier, session timestamp, silo assignment, symbolic output, verification status, noise reduction score, and a synthesis summary. Each block hashes its own content combined with the previous block hash, producing a tamper-evident chain. The full Chronicle is

publicly readable and downloadable. A daily synthesis pass collapses all verified session outputs to a single Chronicle block per day, maintaining a lean and permanently ingestible record.

3.3 The Verification Pipeline

Raw agent outputs enter a staging buffer before Chronicle consideration. Layer one applies symbolic verification via SymPy to detect mathematical redundancy and equivalence with prior entries. Layer two applies a formal verification gateway interfacing with Lean 4 for outputs making mathematical claims. Recent advances in LLM-to-Lean formal verification — including Aristotle achieving gold-medal performance on the 2025 International Mathematical Olympiad with formally verified proofs (arXiv:2510.01346) — confirm that this layer is achievable with current open-source tooling. Only outputs passing both layers enter the Chronicle.

3.4 The Semantic Compression Layer

The Chronicle ingestion layer applies a three-stage compression pipeline to raw agent inputs. Stage one performs structural reduction via JSON normalization. Stage two applies Telegraphic Semantic Compression, retaining high-entropy content tokens while removing grammatical elements reconstructable by the receiving model. Stage three performs a daily synthesis pass. Target compression ratios, based on published benchmarks of component tools, are sufficient to reduce multi-gigabyte raw sessions to Chronicle blocks measurable in kilobytes. Measured performance will be reported following system deployment.

3.5 The Universal Document Ingestion Pipeline

A companion pipeline enables batch processing of archival documents in any format — PDF, image, CSV, Excel, structured and unstructured text. OCR extraction, semantic normalization, noise subtraction, and JSON synthesis produce a single compressed signal file suitable for direct AI ingestion. The pipeline design draws on applied archival research methodology developed independently by the author, including OCR extraction and structured data synthesis from genealogical microfilm records.

3.6 Infrastructure

The proposed system operates entirely on zero-cost open infrastructure. The job queue uses Redis RQ on Upstash free tier. Compute runs on Koyeb permanent free tier with Google Cloud Run as failover. The public Chronicle syncs daily to a Hugging Face Dataset repository. Chain integrity is verified on each commit via GitHub Actions on a public repository, which offers unlimited free execution for public repositories. Major verified milestones receive permanent DOI deposit via Zenodo.

4. The Subtraction Methodology

The operational logic of the Foundry derives from the Presignal Subtraction Framework. Each agent session is evaluated not only for what it contributes positively but for how much of the prior solution space it eliminates. Noise reduction scores are calculated per

session and accumulated in the Contribution Registry. The Chronicle therefore contracts the active problem space with each verified entry, regardless of whether that entry represents a positive result or a confirmed dead end.

This approach treats verified negative results as first-class scientific contributions — a departure from conventional publication models that systematically undervalue null results. The Chronicle's value compounds over time: each verified dead end eliminates a region of solution space permanently, allowing subsequent agents to concentrate exclusively on uneliminated territory.

5. Open Participation and Scaling

The framework is designed to scale from a single operator with three workers to institutional deployments with thousands of parallel workers without architectural modification. External institutions contribute compute by connecting workers to the published queue endpoint. Worker contributions are recorded in the Contribution Registry and attributed in Chronicle milestone blocks. A published scaling protocol document specifies the technical requirements for institutional participation.

The science produced by the Chronicle is fully open. The Chronicle itself is publicly readable and downloadable. The Contribution Registry is public. All architectural documentation is published under Creative Commons CC-BY 4.0. Commercial use of the system's infrastructure and credential gating services is governed by separate agreement with Presignal Inc.

6. Conclusion

The Asymptotic Foundry proposes the first open, verifiable, permanently auditable multi-agent scientific research pipeline combining priority-tiered access control, hash-chained ledger architecture, formal verification gating, and semantic compression into a unified zero-cost deployable framework. The system is designed to run indefinitely without ongoing cost to the operator, accumulating a permanently auditable scientific record that any researcher or institution can download, verify, and build upon.

All components are open source. The Chronicle is publicly readable. The architecture is published under CC-BY 4.0. Full implementation documentation, component specifications, and the scaling protocol will be maintained at the project repository following system deployment.

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Disclosure

This paper describes a proposed architectural framework. The system has not yet been deployed in production form. All performance claims reference published benchmarks of individual component tools. Measured system performance will be reported in subsequent publications following deployment. The author has no conflicts of interest to declare.